

Hardware, Software and OS

the physical parts, and everything that runs on them

Hardware

the physical parts, and how they fit together

What Is Hardware?

DEFINITION

Hardware is the physical part of a computer that we can **see and touch**.

That is the whole test, and it is a genuinely useful one. Keyboard, mouse, monitor, the CPU chip, the RAM stick, the SSD inside your laptop — every part you could physically hold in your hand is hardware. If you cannot touch it, it is software.

Types of Hardware

Hardware is much easier to remember when you group it by **the job it does** rather than by what it looks like. There are four groups.

CATEGORY	EXAMPLES
Input Devices	Keyboard, Mouse, Scanner, Microphone
Output Devices	Monitor, Printer, Speakers
Storage Devices	HDD, SSD, Pen Drive, Memory Card

CATEGORY	EXAMPLES
Internal Components	CPU, GPU, RAM, Power Supply, Motherboard

INTERNAL COMPONENTS, IN DETAIL

CPU

Processes instructions. The brain of the machine.

GPU

Graphics processing — images, video, games.

RAM

Temporary working memory used while programs run.

Power Supply

Provides power to every component.

Motherboard

Connects everything together.



Notice that the first three categories map exactly onto **input**, **output** and **storage** from Chapter 1. The internal components are the **process** part. The definition of a computer is sitting right there inside its own hardware.

Memory — The Proper Types

People use the words "memory" and "storage" as if they mean the same thing. They do not. There are four distinct levels, and they exist because of one trade-off: **fast memory is expensive and small, cheap memory is slow and large**. So a computer uses all four together, each for a different purpose.

TYPE	WHAT IT DOES
Register	The smallest and fastest memory of all — built into the CPU itself, holding the exact data being processed right now.
Cache Memory	Small, very fast memory sitting between the CPU and RAM, holding data the CPU is likely to reuse soon.
Primary Memory	Directly used by the CPU while a program runs (RAM). Temporary — cleared when the power goes off.
Secondary Memory	Long-term storage that keeps data even when the power is off (HDD, SSD).

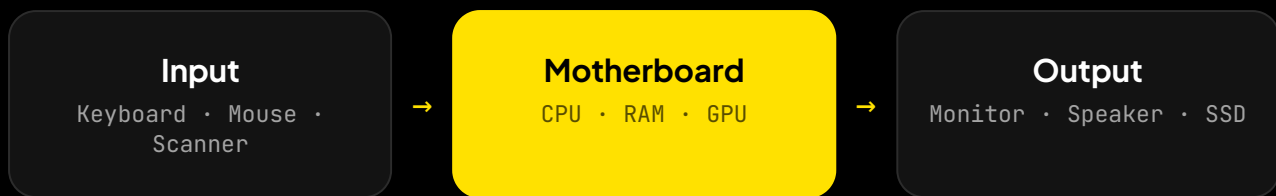
Register → Cache → RAM (Primary) → Storage (Secondary)

fastest, smallest —————→ slowest, largest

 **Common mistake:** assuming RAM and storage are the same thing. RAM is temporary — switch off the power and everything in it is gone. An SSD keeps your data. That is exactly why unsaved work disappears when your laptop suddenly dies.

How the Pieces Connect

These parts do not just sit near each other inside the case — they are all wired together through one board.



The motherboard is the hub that connects every major component. Input devices feed data in, the CPU, RAM and GPU sit on the board and do the work, and the results go out to your monitor, speakers or drive.

SUMMARY

- Hardware is any physical part you can see and touch.
- Four groups: input devices, output devices, storage devices, internal components.
- Memory hierarchy: Register → Cache → RAM → Storage, fastest and smallest to slowest and largest.
- RAM is temporary; SSD and HDD keep data when the power is off.
- The motherboard connects everything.

Software & Operating System

what runs on top of the hardware

Hardware Is Not Enough

Imagine a CPU, some RAM, a keyboard and a monitor sitting on a desk, all connected and powered on. Nothing happens. Something has to tell those parts what to do — and that something is software.

DEFINITION

Software is the set of **programs and instructions** that tell the hardware what to do.

Types of Software

Software is grouped by **who it serves**.

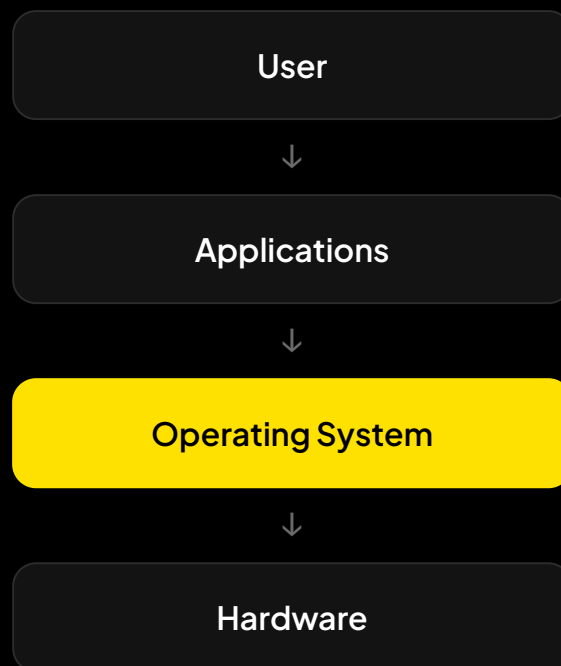
TYPE	EXAMPLES	WHAT IT DOES
System Software	Operating System, Device Drivers	Runs the machine itself
Application Software	Chrome, VLC, MS Word, VS Code	Does a job for you
Utility Software	Antivirus, Disk cleanup tools	Maintains and protects the machine

What Is an Operating System?

DEFINITION

An operating system is **system software that manages computer hardware and provides a platform for applications to run**.

Think of it as a middle layer. You never talk to the hardware directly. You talk to an app, the app talks to the operating system, and the operating system talks to the hardware. When you save a file in Word, Word does not know how your SSD works — it just asks the OS to save it.



Functions of an Operating System

FUNCTION	WHAT IT HANDLES
Process Management	Starts, runs, and switches between programs.
Memory Management	Allocates and frees RAM for running programs.
File Management	Organizes, stores, and controls access to files.
Device Management	Talks to hardware — printers, disks, network.

Operating Systems You Will Run Into

OPERATING SYSTEM	TYPICALLY FOUND ON
Windows	Laptops, Desktop PCs
macOS	MacBook, iMac

OPERATING SYSTEM	TYPICALLY FOUND ON
Linux	Servers, developer machines
Android	Smartphones, tablets
iOS	iPhone, iPad

Every one of these does the same four jobs listed above. What changes is the hardware they run on and the interface you see.

SUMMARY

- Software is the instructions that tell hardware what to do.
- Three types: system, application, utility.
- The OS manages hardware and gives applications a platform to run on.
- Four OS functions: process, memory, file and device management.